

DESIGN 01

Class Diagram of an Online Sales System

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Overview

Figure 1 presents a UML class diagram for an online shopping system. It separates the design into a **control** package, which contains controllers and data access objects (DAOs), and a **model** package, which contains business entities. The main functions cover customers, products, shopping carts, orders, shipping, and payments.

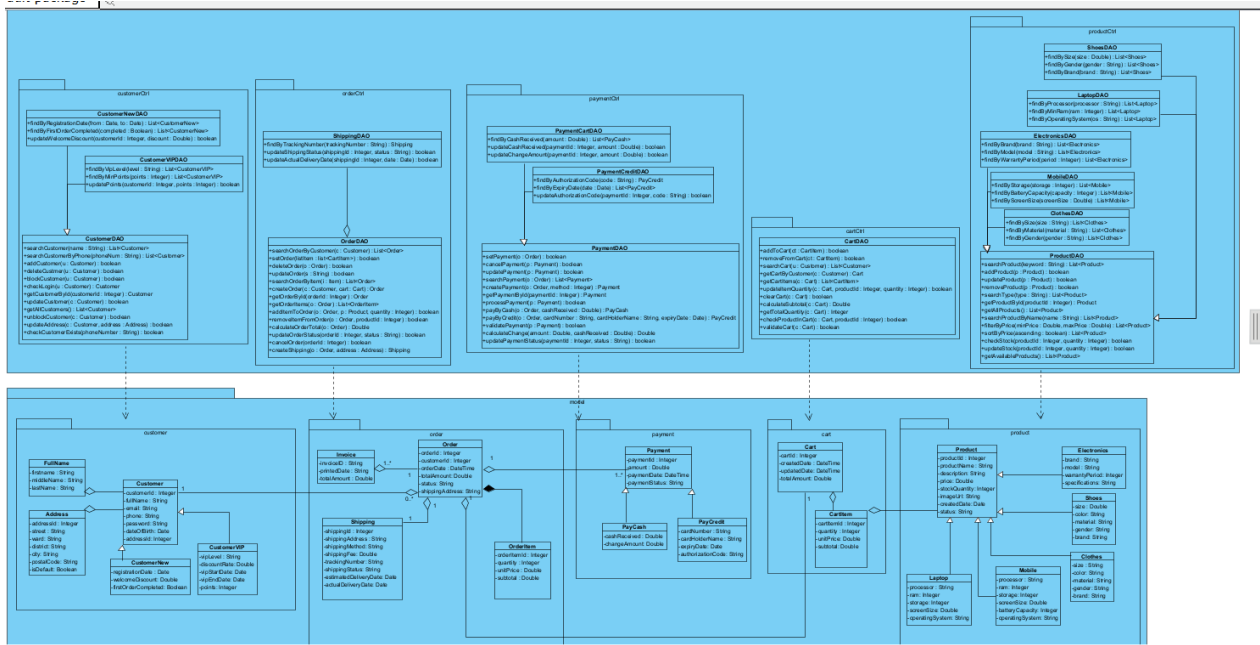


Figure 1: Class diagram of the online shopping system.

Brief Description

The customer group stores customer details and distinguishes **CustomerVip** and **CustomerNew** from the base **Customer** class. The product group defines a common **Product** class and specialized types such as laptops, mobile phones, shoes, and clothes. A **Cart** contains **CartItem** objects that record selected products and quantities.

When a customer places an order, the system creates an **Order** with **OrderItem** records. Related classes represent the invoice and shipping information. The **Payment** class has two specialized forms, **PayCash** and **PayCredit**. In the control package, the corresponding DAO classes handle operations such as searching products, updating the cart, processing orders, and recording payments.

The diagram uses inheritance for specialized customers, products, and payment methods. Associations connect customers, carts, orders, items, and products. Overall, the design groups business data and data access responsibilities by function.